Math 231B HW 2

Zih-Yu Hsieh

February 05, 2026

1 (iv) ND :(

Problem 1

Let Q be the root lattice of a simple Lie algebra $\mathfrak{g}$, Q_+ be its positive part. Define the **Kostant partition function** to be the function $p : Q \to \mathbb{Z}_{\geq 0}$ which attaches to $\beta \in Q_+$ the number of ways to write β as a sum of positive roots of $\mathfrak{g}$ (where the order does not matter), and $p(\beta) = 0$ if $\beta \notin Q_+$.

(i) Show that

$$\sum_{\beta \in Q_+} p(\beta) e^{-\beta} = \frac{1}{\prod_{\alpha \in R_+} (1 - e^{-\alpha})} \tag{1.1}$$

(ii) Prove the **Kostant multiplicity formula**

$$\dim L_\lambda[\gamma] = \sum_{w \in W} (-1)^{\ell(w)} p(w(\lambda + \rho) - \rho - \gamma) \tag{1.2}$$

(iii) Compute $p(k_1\alpha_1 + k_2\alpha_2)$ for $\mathfrak{g} = \mathfrak{sl}_3$ and $\mathfrak{g} = \mathfrak{sp}_4$.

(iv) Use (iii) to compute explicitly the weight multiplicities of the irreducible representations L_λ for $\mathfrak{g} = \mathfrak{sl}_3$ and $\mathfrak{g} = \mathfrak{sp}_4$. (You should get a sum of 6, respectively 8 terms, not particularly appealing, but easily coputable in each special cases).

Solution: Here, we'll assume $\alpha_1, ..., \alpha_n$ are all the positive roots under some polarization.

(i) First, suppose $\beta \in Q_+$, then $\beta = k_1\alpha_1 + ... + k_n\alpha_n$ for some $k_1, ..., k_n \in \mathbb{N}$. As a result:

$$e^{-\beta} = e^{-k_1\alpha_1 - ... - k_n\alpha_n} = e^{-k_1\alpha_1} ... e^{-k_n\alpha_n} \tag{1.3}$$

Now, consider the fact that as operators, all $\alpha \in R_+$ satisfies the following:

$$\frac{1}{1 - e^{-\alpha}} = \sum_{k=0}^{\infty} e^{-k\alpha} \tag{1.4}$$

Hence, the product formula (after expanding the product) becomes the following:

$$\frac{1}{\prod_{\alpha \in R_+} (1 - e^{-\alpha})} = \prod_{\alpha \in R_+} \left(\sum_{k=0}^{\infty} e^{-k\alpha} \right) = \sum_{(k_1, ..., k_n) \in \mathbb{N}^n} e^{-k_1\alpha_1 - ... - k_n\alpha_n} \tag{1.5}$$

Notice that each $(k_1, ..., k_n) \in \mathbb{N}^n$ corresponds to a unique $\beta \in Q_+$, namely $\beta = k_1\alpha_1 + ... + k_n\alpha_n$. And, the number of elements in $\mathbb{N}^n$ corresponding to each $\beta \in Q_+$ is precisely given by the partition $p(\beta)$ (i.e. distinct numbers of nonnegative integer combinations of simple roots that form β, which corresponds to the distinct n-tuples corresponding to β).

Hence, the sum is naturally equal to $\sum_{\beta\in Q_+} p(\beta)e^{-\beta}$, since there are $p(\beta)$ distinct n-tuples $(k_1, ..., k_n) \in \mathbb{N}^n$ satisfying $\beta = k_1\alpha_1 + ... + k_n\alpha_n$.

(ii) Using the Weyl Character Formulas, one gets the following characterization:

$$\begin{aligned}\chi_\lambda &= \frac{\sum_{w\in W}(-1)^{\ell(w)}e^{w(\lambda+\rho)}}{\prod_{\alpha\in R_+}(e^{\frac{\alpha}{2}} - e^{-\frac{\alpha}{2}})} \\ &= \frac{\sum_{w\in W}(-1)^{\ell(w)}e^{w(\lambda+\rho)}}{\prod_{\alpha\in R_+}(1 - e^{-\alpha})e^{\frac{\alpha}{2}}} \\ &= \frac{\sum_{w\in W}(-1)^{\ell(w)}e^{w(\lambda+\rho)}e^{-\rho}}{\prod_{\alpha\in R_+}(1 - e^{-\alpha})} \\ &= \left(\sum_{w\in W}(-1)^{\ell(w)}e^{w(\lambda+\rho)-\rho}\right)\prod_{\alpha\in R_+}(1 + e^{-\alpha} + e^{-2\alpha} + ...)\end{aligned} \tag{1.6}$$

(Note: recall the definition $\rho := \frac{1}{2}\sum_{\alpha\in R_+}\alpha$).

$$\chi_\lambda = \sum_{\gamma\in P}\dim L_\lambda[\gamma]e^\gamma \tag{1.7}$$

Given any $\gamma \in P$, suppose the n-tuples $(k_1, ..., k_n) \in \mathbb{N}^n$ (corresponding to $\beta = k_1\alpha_1 + ... + k_n\alpha_n$) satisfies $e^\gamma = e^{w(\lambda+\rho)-\beta-\rho}$, one has $\gamma = w(\lambda + \rho) - \beta - \rho$, or $\beta = w(\lambda + \rho) - \gamma - \rho$ (here, denote it as $\beta_{w,\gamma}$). Based on part (i), one knows there are total of $p(\beta_{w,\gamma})$ number of $e^{-\beta_{w,\gamma}}$ appearing in the large product above, then the term can also be rewritten as the following:

$$\begin{aligned}\chi_\lambda &= \left(\sum_{w\in W}(-1)^{\ell(w)}e^{w(\lambda+\rho)-\rho}\right)\prod_{\alpha\in R_+}(1 + e^{-\alpha} + e^{-2\alpha} + ...) \\ &= \sum_{\gamma\in P}\left(\sum_{w\in W}(-1)^{\ell(w)}p(\beta_{w,\gamma})e^{w(\lambda+\rho)-\beta_{w,\gamma}-\rho}\right) \\ &= \sum_{\gamma\in P}\left(\sum_{w\in W}(-1)^{\ell(w)}p(\beta_{w,\gamma})e^{\gamma}\right)\end{aligned} \tag{1.8}$$

So, the coefficient of e^γ must be $\sum_{w\in W}(-1)^{\ell(w)}p(\beta_{w,\gamma})$, where $p(\beta_{w,\gamma}) = p(w(\lambda + \rho) - \gamma - \rho)$. And, the second equation states that such coefficient is $\dim L_\lambda[\gamma]$. So, one gets:

$$\dim L_\lambda[\gamma] = \sum_{w\in W}(-1)^{\ell(w)}p(w(\lambda + \rho) - \gamma - \rho) \tag{1.9}$$

(iii) **Case for $\mathfrak{sl}_3$:**

First, for the case of $\mathfrak{sl}_3$ (or A_2, with 6 roots), since the graphs are formed by structure similar to regular hexagons, let α_1, α_2 be two simple roots under some polarization, then the third positive root $\alpha_3 = \alpha_1 + \alpha_2$.

Now, given arbitrary $k_1, k_2 \in \mathbb{Z}$, to consider $p(k_1\alpha_1 + k_2\alpha_2)$: Since if $k_1 < 0$ or $k_2 < 0$, $k_1\alpha_1 + k_2\alpha_2 \notin Q_+$, then $p(k_1\alpha_1 + k_2\alpha_2) = 0$. Now, we'll focuse on $k_1, k_2 \geq 0$.

To consider the case $k_1 \leq k_2$, notice that if $l_1, l_2, l_3 \in \mathbb{N}$ satisfies $l_1\alpha_1 + l_2\alpha_2 + l_3\alpha_3 = k_1\alpha_1 + k_2\alpha_2$, then one has the following:

$$(l_1 + l_3)\alpha_1 + (l_2 + l_3)\alpha_2 = k_1\alpha_1 + k_2\alpha_2 \tag{1.10}$$

Which, since the simple roots α_1, α_2 form a basis, the above enforces $l_1 + l_3 = k_1$ and $l_2 + l_3 = k_2$. With all components being nonnegative together with $k_1 \leq k_2$, one observes that $0 \leq l_3 \leq k_1 \leq k_2$, which implies there are at most $k_1 + 1$ distinct choices for l_3 (namely $0, 1, ..., k_1$). However, for each fixed $l_3 \in \{0, 1, ..., k_1\}$, one has the equation $l_1 = k_1 - l_3 \geq 0$ and $l_2 = k_2 - l_3 \geq 0$ by how we define k_1, k_2, and l_3. So, each fixed l_3 corresponds to a unique tuple, showing there are total of $k_1 + 1$ ways of writing $k_1\alpha_1 + k_2\alpha_2$ ($k_1 \leq k_2$) as nonnegative integer sums of the positive roots $\alpha_1, \alpha_2, \alpha_3 = \alpha_1 + \alpha_2$ of $\mathfrak{sl}_3$.

Finally, for the case $k_1 > k_2$, notice that if reversing all the inequalities beforehand, one gets that the multiplicity will be $k_2 + 1$ instead. Hence, one can conclude that for $\mathfrak{sl}_3$, the partition function is given as follow:

$$p(k_1\alpha_1 + k_2\alpha_2) = \begin{cases} 0 & k_1 < 0 \text{ or } k_2 < 0 \\ \min\{k_1, k_2\} + 1 & k_1, k_2 \geq 0 \end{cases} \tag{1.11}$$

For the case of $\mathfrak{sp}_4$ (or C_2), let α_1 indicates the shorter simple root and α_2 indicates the longer simple root. Then, the remaining positive roots have $\alpha_3 = \alpha_1 + \alpha_2$, and $\alpha_4 = 2\alpha_1 + \alpha_2$.

Similarly, given arbitrary $k_1\alpha_1 + k_2\alpha_2$ (where $k_1, k_2 \in \mathbb{Z}$), if $k_1 < 0$ or $k_2 < 0$, since $k_1\alpha_1 + k_2\alpha_2 \notin Q_+$, one has $p(k_1\alpha_1 + k_2\alpha_2) = 0$. So, we'll again focus on the case $k_1, k_2 \geq 0$.

First, look at the case $k_1 \leq 2k_2$: If the integers $l_1, l_2, l_3, l_4 \in \mathbb{N}$ satisfies $k_1\alpha_1 + k_2\alpha_2 = l_1\alpha_1 + l_2\alpha_2 + l_3\alpha_3 + l_4\alpha_4$, one gets the following simplification:

$$(l_1 + l_3 + 2l_4)\alpha_1 + (l_2 + l_3 + l_4)\alpha_2 = k_1\alpha_1 + k_2\alpha_2 \tag{1.12}$$

With α_1, α_2 being a basis, this enforces $l_1 + l_3 + 2l_4 = k_1$ and $l_2 + l_3 + l_4 = k_2$. So, the first requirement is: $0 \leq 2l_4 \leq k_1$ and $0 \leq l_4 \leq k_2$, or $0 \leq l_4 \leq \frac{k_1}{2} \leq k_2$. This enforces $0 \leq l_4 \leq \left\lfloor \frac{k_1}{2} \right\rfloor$ (so, there are at most $\left\lfloor \frac{k_2}{2} \right\rfloor + 1$ choices for l_4, namely $0, 1, ..., \left\lfloor \frac{k_1}{2} \right\rfloor$).

Then, when fixing l_4, the system reduces to $l_1 + l_3 = k_1 - 2l_4$ and $l_2 + l_3 = k_2 - l_4$. Notice that with $2l_4 \leq k_1$ and $l_4 \leq k_2$, the above has $0 \leq k_1 - 2l_4, k_2 - l_4$. So, again with $l_1, l_2, l_3 \in \mathbb{N}$, it reduces to the case computed for $\mathfrak{sl}_3$, namely there are $\min\{k_1 - 2l_4, k_2 - l_4\} + 1$ distinct choices of the 3-tuple (l_1, l_2, l_3). Hence, the following is the total number of ways (given $k_1 \leq k_2$):

$$\sum_{l_4=0}^{\left\lfloor \frac{k_1}{2} \right\rfloor} \min\{k_1 - 2l_4, k_2 - l_4\} + 1 \tag{1.13}$$

There are two separated case to consider now: If $k_1 \leq k_2$ is true, then $k_1 - 2l_4 \leq k_2 - l_4$ for all given l_4. Hence, the equation reduces to the following:

$$\begin{aligned}\sum_{l_4=0}^{\lfloor \frac{k_1}{2} \rfloor} \min\{k_1 - 2l_4, k_2 - l_4\} + 1 &= \sum_{l_4=0}^{\lfloor \frac{k_1}{2} \rfloor} (k_1 - 2l_4 + 1) \\ &= \left(\left\lfloor \frac{k_1}{2} \right\rfloor + 1\right)(k_1 + 1) - \left\lfloor \frac{k_1}{2} \right\rfloor \left(\left\lfloor \frac{k_1}{2} \right\rfloor + 1\right) \\ &= \left(\left\lfloor \frac{k_1}{2} \right\rfloor + 1\right)\left(k_1 - \left\lfloor \frac{k_1}{2} \right\rfloor + 1\right)\end{aligned} \tag{1.14}$$

Else, if $k_1 > k_2$, we aim to find the range where $k_1 - 2l_4 \leq k_2 - l_4$, or when $k_1 \leq k_2 + l_4$, equivalent to say $k_1 - k_2 \leq l_4$. Notice that by our initial assumption, $k_1 \leq 2k_2$, then one has $k_1 - k_2 \leq k_1 - \frac{k_1}{2} = \frac{k_1}{2}$, so $k_1 - k_2 \leq \left\lfloor \frac{k_1}{2} \right\rfloor$ (since both are integers with the same bound), hence it makes sense to talk about $k_1 - k_2 \leq l_4$. As a result the formula becomes:

$$\begin{aligned}\sum_{l_4=0}^{\lfloor \frac{k_1}{2} \rfloor} \min\{k_1 - 2l_4, k_2 - l_4\} + 1 &= \sum_{l_4=0}^{k_1-k_2-1} (k_2 - l_4 + 1) + \sum_{l_4=k_1-k_2}^{\lfloor \frac{k_1}{2} \rfloor} (k_1 - 2l_4 + 1) \\ &= k_2(k_1 - k_2) + (k_1 + 1)\left(\left\lfloor \frac{k_1}{2} \right\rfloor - k_1 + k_2\right) \\ &\quad - \left\lfloor \frac{k_1}{2} \right\rfloor \left(\left\lfloor \frac{k_1}{2} \right\rfloor + 1\right) + (k_1 - k_2 - 1)\frac{k_1 - k_2}{2} \\ &= (k_1 + k_2 - 1)\frac{k_1 - k_2}{2} + (k_1 + 1)\left(\left\lfloor \frac{k_1}{2} \right\rfloor - k_1 + k_2\right) \\ &\quad - \left\lfloor \frac{k_1}{2} \right\rfloor \left(\left\lfloor \frac{k_1}{2} \right\rfloor + 1\right)\end{aligned} \tag{1.15}$$

Now, to consider the case $k_1 > 2k_2$, then the previous restrictions become $0 \leq l_4 \leq \frac{k_1}{2}$ and $0 \leq l_4 \leq k_2$, in this case $k_2 < \frac{k_1}{2}$, so the bound becomes $0 \leq l_4 \leq k_2$ (so $l_4 = 0, 1, ..., k_2$).

Then, with any fixed l_4, one requires $l_1 + l_3 = k_1 - 2l_4$, and $l_2 + l_3 = k_2 - l_4$; again using the case computed for $\mathfrak{sl}_3$, one has total of $\min\{k_1 - 2l_4, k_2 - l_4\} + 1$ different ways of expressing $(l_1, l_2, l_3) \in \mathbb{N}^3$. So, the formula is the following:

$$\sum_{l_4=0}^{k_2} \min\{k_1 - 2l_4, k_2 - l_4\} + 1 \tag{1.16}$$

Which, since $k_2 - l_4 \leq 2k_2 - 2l_4 < k_1 - 2l_4$, one has the minimum be $k_2 - l_4$ always. SO, the equation reduces to the following:

$$\begin{aligned}\sum_{l_4=0}^{k_2} \min\{k_1 - 2l_4, k_2 - l_4\} + 1 &= \sum_{l_4=0}^{k_2} (k_2 - l_4 + 1) \\ &= (k_2 + 1)^2 - \frac{k_2(k_2 + 1)}{2} \\ &= \frac{(k_2 + 1)(k_2 + 2)}{2}\end{aligned} \tag{1.17}$$

Finally, here's the full list:

$$p(k_1\alpha_1 + k_2\alpha_2) = \begin{cases} 0 & k_1 < 0 \text{ or } k_2 < 0 \\ \left(\left\lfloor \frac{k_1}{2} \right\rfloor + 1\right)\left(k_1 - \left\lfloor \frac{k_1}{2} \right\rfloor + 1\right) & 0 \leq k_1 \leq k_2 \\ \frac{(k_1+k_2-1)(k_1-k_2)}{2} + (k_1+1)\left(\left\lfloor \frac{k_1}{2} \right\rfloor - k_1 + k_2\right) & \\ -\left\lfloor \frac{k_1}{2} \right\rfloor\left(\left\lfloor \frac{k_1}{2} \right\rfloor + 1\right) & k_2 < k_1 \leq 2k_2 \\ \frac{(k_2+1)(k_2+2)}{2} & 2k_2 < k_1 \end{cases} \tag{1.18}$$

(iv) Didn't finish the calculation :(